

SUMUKH ACHARYA

+91 9972454072 | sumukh.acharya@gmail.com

[linkedin.com/in/sumukhacharya](https://www.linkedin.com/in/sumukhacharya) | github.com/sumukhacharya03 | sumukh-acharya.vercel.app

SKILLS

Languages: Python, JavaScript, MySQL

Backend & Tools: FastAPI, Flask, Docker, Git, Kafka

ML Frameworks & Tools: SLM, PyTorch, Transformers, TRL, Unsloth, HuggingFace, Ollama

ML Techniques: LoRA/QLoRA, DPO, SSM/Mamba, RVQ

EXPERIENCE

Machine Learning Engineer

tvam Technologies

Jul 2026 - Present

Bengaluru, India

- Continuing DPO fine-tuning of the SLM, currently refining edge-case handling; results and evaluations to follow.
- Continuing integration of the audio codec and text-to-audio-token models into a unified speech-to-speech (STS) pipeline; results and evaluations to follow.

Machine Learning Intern

tvam Technologies

Sep 2025 - Jun 2026

Bengaluru, India

Agentic Tool-Calling SLM | LoRA/QLoRA, Instruction-Tuned Fine-Tuning

- Fine-tuned SLM for agentic tool-calling through 5 iterative cycles of instruction tuning, edge-case diagnosis, and synthetic data regeneration using larger LLMs, improving correct tool-call accuracy by 85% over the base model.

Conversational Fine-Tuning for Human-Like Interaction | LoRA/QLoRA, SFT

- Self-curated a 1.5M-token conversational dataset from scratch by sourcing and manually curating interview transcripts from a couple of public figures into a JSONL corpus, then supervised fine-tuned the tool-calling SLM for conversational quality and emotional nuance, narrowing the performance gap to GPT-5.2 by 75% on human-evaluated conversational scenarios.

DPO Alignment for Emotional Maturity | DPO, SFT

- Fine-Tuning the latest fine-tuned SLM on 14M DPO preference pairs curated from LLM-generated conversation logs, using a hybrid DPO + SFT objective to jointly optimize preference alignment and next-token prediction, with edge-case testing and loss optimization currently in progress.

Audio Codec & Text-to-Audio-Token Modeling | PyTorch, torchaudio, GAN, BiMamba2

- Training a GAN-based neural audio codec for high-fidelity audio reconstruction, currently at 3,750 epochs with near-lossless reconstruction quality on most inputs and a handful of edge cases still being refined, alongside a separate BiMamba2-based neural network to convert text into audio tokens – the two core components for an eventual speech-to-speech (STS) system.

Teaching Assistant

PES University | Python, Databases, Linear Algebra

Aug 2025 - May 2026

Bengaluru, India

- Mentored 250+ students combined across Python, Databases, and Linear Algebra through lab sessions, assignment grading, and one-on-one doubt-clearing and debugging sessions, while preparing lab material for each subject.

Data Science Intern

CODMAV Research Centre (PES University)

Jun 2024 - Aug 2024

Bengaluru, India

- Built a [lung cancer risk prediction system](#) on 22,811 patient records, achieving 98.75% accuracy and 96.25% recall using a soft-voting ensemble of XGBoost, CatBoost, SVM, and KNN on RFE-selected features.

- Engineered a multi-stage data pipeline reducing 788 clinical features to 89 medically-validated markers via RF imputation and data filtering.
- Compared 4 feature selection algorithms (PCA, BSO, RFE, SelectKBest) across multiple model architectures to identify the optimal clinical feature subset.
- Published findings at IEEE InC4 2025, indexed in IEEE Xplore.

PROJECTS

Grid Pulse — Real-Time F1 Stock Market | Python, Apache Kafka, Streamlit

- Built a real-time data pipeline simulating a stock market for F1 drivers, processing both historical batch data and live race weekend streams via Kafka.
- Designed a distributed architecture decoupling data generation, processing, and visualization with cron-triggered real-time updates and a Streamlit dashboard.

DFOS — Distributed File Orchestration and Synchronization | Python, Sockets, Threading

- Built a multi-client file transfer system with concurrent client handling using ThreadPoolExecutor, supporting upload, download, preview, and delete operations.
- Implemented user authentication, dedicated per-user storage, and performance logging with CPU/memory monitoring on shutdown.

PUBLICATIONS

Acharya, S., et al. **Predictive Analytics for Early Lung Cancer Risk using Machine Learning.** IEEE International Conference on Contemporary Computing and Communications (InC4), 2025. Indexed in IEEE Xplore.

ACHIEVEMENTS & CERTIFICATIONS

- Ranked **7th out of 364 teams** in [Kaggle Data Analytics Hackathon](#).
- [IBM Data Science Professional Certificate](#) — Coursera.

EDUCATION

B.Tech in Computer Science and Engineering

PES University, Bengaluru | Specialisation: Machine Intelligence and Data Science

Sep 2022 - May 2026

CGPA: 7.7